Knowing When Not to Answer: Abstention Drift in Multi-Turn Retrieval-Augmented Conversations

CS683 Natural Language Processing • Initial Project Report

Team: *Member 1* • *Member 2* • *Member 3* • *Member 4* • *Member 5*

1 Problem Statement

Retrieval-augmented generation (RAG) assistants are widely deployed as conversational agents, but they often answer confidently even when the retrieved documents do not contain the answer. In the SemEval-2026 multi-turn RAG task (MTRAGEval), one of the strongest systems reached 83.9% answerability accuracy yet detected only **21.8% of unanswerable turns** (development set) [3]: its LLM judges favoured plausible-sounding answers over evidence. Research on abstention (a model declining to answer) is almost entirely single-turn [4, 5], while real conversations get longer and more context-dependent as they go on. We ask:

RQ1. Does detection of unanswerable questions get worse at later turns in a conversation (*abstention drift*)?

RQ2. Can a lightweight answerability checker, trained on cheaply generated unanswerable examples, reduce this drift *without* increasing wrong refusals on answerable turns?

2 Datasets

We use the **MTRAG** benchmark family (Apache-2.0), which also serves as the data for SemEval-2026 Task 8.

MTRAG [1] (TACL 2025): 110 human-written conversations (avg. 7.7 turns) converted to 842 tasks. Each task is one turn with its full history, a reference answer, human-labelled reference passages, and an answerability label (709 answerable, 68 partial, 55 unanswerable, 10 conversational). There are four domains: ClapNQ (Wikipedia), Cloud (technical documentation), FiQA (finance forums) and Govt (government web pages). Their passage-level corpora total about 366K passages. Human-written standalone rewrites of each question are also provided, and we use them as an upper bound for query rewriting. *Use:* train/dev, split 80/20 *by conversation* to avoid leakage.

MTRAG-UN [2] (Findings ACL 2026): the SemEval-2026 evaluation benchmark, focused on UNanswerable, UNderspecified and non-standalone questions. The 507 publicly released tasks (285 answerable, 47 partial, 97 unanswerable, 78 underspecified) use the same four corpora. *Use:* held-out test set.

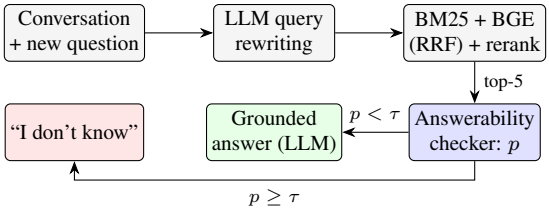
% unanswerable	Turns 1–2	Turns 3–5	Turns 6+
MTRAG (842 tasks)	5.0	6.8	7.4
MTRAG-UN (507 tasks)	9.4	13.8	33.7

Table 1: Share of unanswerable tasks by turn position (our analysis of the released data).

Table 1 shows that unanswerable questions cluster in late turns. A raw comparison across turns is therefore *confounded*, and our method controls for this (§3.3).

3 Proposed Method

3.1 Pipeline



1. **Query rewriting:** an instruction-tuned LLM (Qwen2.5-3B/7B, 4-bit) turns each follow-up question into a standalone query.
2. **Hybrid retrieval:** BM25 and dense BGE-small [7] results are merged with reciprocal rank fusion [6], then re-ranked by a BGE cross-encoder (top-5 passages).
3. **Answerability checker (our contribution):** a DeBERTa-v3-base [8] cross-encoder reads the query and the top-5 passages and outputs $p = P(\text{unanswerable})$. If $p \geq \tau$ the system replies "I don't know"; otherwise the LLM answers only from the passages.

3.2 Evidence removal: free unanswerable data

Only 6.5% of MTRAG turns are unanswerable, too few to train on. For every *answerable* turn we know which passages hold the answer. Re-running retrieval while excluding *all* passages from those source documents gives a realistic context that no longer supports the answer, producing a new unanswerable example at no annotation cost. Answerable examples are also built from retrieval (with the gold passages inserted), so the checker cannot learn a "gold passages look different" shortcut. We drop generated examples whose passages still cover the reference answer (a word-overlap filter plus an optional LLM check), and we hand-check 50 kept examples to report label noise. The threshold τ maximises abstention F1 subject to refusing $\leq$5% of answerable dev turns (RQ2).

3.3 Evaluation

Abstention quality: precision, recall and F1 on unanswerable turns, and the false-abstention rate on answerable turns, per turn bucket (1–2, 3–5, 6+). **Abstention drift** is the slope β of $P(\text{abstain} \mid \text{unanswerable})$ against turn number t:

$$\Pr[\hat{a}_i = 1] = \alpha + \beta\, t_i, \qquad y_i = \text{UNANSWERABLE}$$

with a 95% CI from a bootstrap that resamples whole conversations. To separate turn position from question type, we (i) fit a logistic regression controlling for domain and follow-up type, and (ii) run a *synthetic probe*: evidence removal applied at every turn position of the dev conversations, so only the passages change. **Answer quality:** official MTRAG metrics (ROUGE-L, BERTScore, LLM-judge faithfulness).

Systems compared: S0 last-turn query + BM25 + prompt-only refusal; S1 + LLM rewriting; S2 + hybrid retrieval and re-ranking; S3 = S2 + checker trained on real labels only; **S4** = S2 + checker trained on real + synthetic data (proposed); S4-oracle uses human rewrites (upper bound).

Member	Role	Responsibilities	Key deliverable (week)
Member 1	Retrieval lead	Corpus indexing (BM25, BGE, FAISS), reciprocal rank fusion, cross-encoder re-ranking, retrieval recall@5/nDCG, retrieval ablations	Indexes for 4 domains (W1); S1–S2 retrieval (W2)
Member 2	Generation & rewriting	LLM query rewriting, grounded-answer and refusal prompts, running S0–S4, official MTRAG answer-quality evaluation	S0 baseline (W1); S0–S4 predictions (W3)
Member 3	Checker & data	Evidence-removal synthesis and filters, DeBERTa-v3 checker training, threshold selection under the false-abstention budget	Synthetic data (W2); trained S3/S4 checkers (W2–W3)
Member 4	Evaluation & analysis	Per-turn metrics, drift slope with bootstrap CIs, controlled regression, synthetic probe, plots, error analysis	Evaluation script (W1); results tables and plots (W3–W4)
Member 5	Validation, demo & documentation	Novelty check of SemEval-2026 papers, 50-example human check of synthetic labels, Gradio demo, README, report and slide integration	Novelty memo (W1); demo (W3); report and slides (W4)

Table 2: Division of work. All members contribute to the final report and presentation.

4 Novelty over Prior Work

MTRAG and MTRAG-UN [1, 2] report unanswerable performance as a single aggregate number, and SemEval-2026 systems focus on leaderboard scores [3]. Abstention studies [4, 5] evaluate isolated questions. To our knowledge, no prior work (i) measures how abstention changes with *turn position* while controlling for question type, or (ii) creates unanswerable training data for conversational RAG by removing evidence documents. Both are cheap to run, reusable, and directly relevant to deployed assistants.

5 Deliverables

- Public GitHub repository: modular code, configuration files, unit tests, reproducible scripts (a starter version with 14 passing tests exists).
- Trained answerability checker, prediction files for S0–S4 in the official MTRAG format, and a hand-checked sample of 50 synthetic examples.
- Results: per-turn abstention tables, drift slopes with CIs, a drift plot, ablations and error analysis.
- Interactive Gradio demo that shows sources and a $P(\text{unanswerable})$ meter.
- Final 2-page report and presentation slides; if results are strong, an extended short paper for an ACL Rolling Review cycle.

6 Division of Work (5 members, 4 weeks)

Each member owns one component end to end and one week-by-week deliverable (Table 2); every member reviews one other member's code through pull requests.

Timeline. W1: data, indexes, S0 baseline, evaluation script, novelty check. W2: rewriting and hybrid retrieval (S1–S2), synthetic data, checker training (S3–S4). W3: MTRAG-UN runs, drift analysis, synthetic probe, demo. W4: error analysis, final report and slides.

7 Risks and Mitigation

- **Prior work on per-turn abstention appears:** checked in week 1. Fallback: study the acceptance bias of LLM answerability judges [3] with the same pipeline.
- **Noisy synthetic labels** (another document still answers): lexical and LLM filters, a 50-example human check, and the threshold tuned only on real dev data.
- **Few real unanswerables** (55 + 97): conversation-level bootstrap CIs, plus the synthetic probe for statistical power.
- **Limited compute:** 4-bit 3B model locally (GTX 1650, 4 GB); final 7B runs on free Kaggle/Colab GPUs.

8 Expected Outcomes

(i) A quantified answer to RQ1: whether, and by how much, abstention recall drops per extra turn, with the confound from question position controlled. (ii) Evidence for RQ2: whether synthetic evidence-removed data improves unanswerable recall over prompt-only refusal (S2) and real-label training (S3) at a fixed 5% false-abstention budget. (iii) A reusable open-source toolkit and demo.

References

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[6] G. Cormack, C. Clarke, S. Büttcher. Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods. *SIGIR*, 2009.

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